



PHYSICS

Where does the last term come from in the two-lens formula: $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$?

Asked 7 years, 1 month ago Modified 4 years, 3 months ago Viewed 3k times



I can derive the formula for lenses in contact: i.e.,

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$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}. \quad (1)$$



But for two lenses separated by a distance d I can't seem to get:



$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}. \quad (2)$$



For the first derivation I let the image distance for the first lens v_1 equal the negative object distance for the second lens. ie. $v_1 = -u_2$

When trying to derive the second formula I let: $u_2 = d - v_1$, and then followed the same procedure. I ended up with a very tedious algebraic expression. Did I go wrong with $u_2 = d - v_1$? How do you derive the second formula?

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edited Sep 11, 2017 at 13:35



Qmechanic ♦

210k 48 575 2.3k

asked Sep 11, 2017 at 10:44



Kantura

1,349 1 14 31

1 Related physics.stackexchange.com/q/247617 – Farcher Sep 11, 2017 at 12:45

1 I think your difficulty lies with from where you measure f and indeed u and v for the *system*. You have no optical centre as ordinarily defined. You are into 'thick lens theory'! This isn't especially difficult (if you're not worried about aberrations) but needs learning! – Philip Wood Sep 11, 2017 at 13:43

2 Answers

Sorted by: Highest score (default)



I can't resist to show how this is done using ray transfer matrices. There are two key parameter at any point along a light ray: the distance x of the point from the optical axis, and

- 7 the angle θ of the ray with the horizontal. Then any optical component of the system can be represented as a 2×2 matrix which transform the pair (x, θ) for the incoming ray into the pair (x', θ') for the outgoing ray:



$$\begin{pmatrix} x' \\ \theta' \end{pmatrix} = \begin{pmatrix} \times & \times \\ \times & \times \end{pmatrix} \begin{pmatrix} x \\ \theta \end{pmatrix}.$$

By using what you know, you can easily write down that

- a thin lens has a matrix

$$L = \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix}$$

where f is the focal distance;

- a thickness d of empty space has a matrix

$$S = \begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}.$$

Onto your problem: we have a lens of focal f_1 (matrix L_1), empty space and another lens of focal f_2 (matrix L_2), so the matrix for the whole system is simply the product of the matrices, in reverse order

$$\begin{aligned} M &= L_2 S L_1 = \begin{pmatrix} 1 & 0 \\ -\frac{1}{f_2} & 1 \end{pmatrix} \begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{1}{f_1} & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 - \frac{d}{f_1} & d \\ -\left(\frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}\right) & 1 - \frac{d}{f_2} \end{pmatrix} \end{aligned}$$

and you can read the focal distance in the bottom left corner! As you see, 99% of my exposition is just explaining the method. The actual computation is that trivial, systematic, triple matrix product. No monkeying about with geometry all the time: you just need to do it once to deduce the matrices for common components.

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edited Sep 11, 2017 at 13:39

answered Sep 11, 2017 at 13:30

user154997

1 The fact that the element M_{11} and $M_{22} \neq 1$ and $M_{12} \neq 0$ implies that the effective lens could not be regarded as a thin lens. – [user279043](#) Mar 22, 2018 at 7:21

@user279043 indeed, the combination of two thin lenses is **not a thin** lens itself (unless they are in contact $d = 0$) [physics.stackexchange.com/questions/536302/...](https://physics.stackexchange.com/questions/536302/) – [BrainOverflow](#) Jul 22, 2022 at 8:30

